The Riemann Kernel Is Globally PF_4

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Abstract

Let Φ be the classical Riemann, or de Bruijn–Newman, kernel. We prove that its exact global Pólya-frequency order is four: every translation minor of order at most four is strictly positive at strictly ordered nodes, whereas the order-five minor at the exact spacing 211/2000 is negative. For the positive half, a two-mode argument proves global positivity of the log-curvature quantities q, F_2 and of the fully confluent invariant $\mathcal{C}_4$. The finite verification uses only rational polynomial algebra, Taylor inequalities, and geometric tails. The direct finite-witness certificate likewise uses exact rational arithmetic. A quotient-Wronskian argument reduces arbitrary fourth-order minors to $\partial_\xi \Psi < 0$. In the increasing curvature coordinate, its numerator is

$$\delta\Lambda \int_p^w \Delta(t) \frac{\mathcal{C}_4(t)}{Q(t)^6 \kappa(t)^2}\, \mathrm{d}t.$$

Here Δ is a deterministic closed cumulative gap built from two explicitly normalized, curvature-tilted triangular weights. Its strict positivity follows on the coordinate interval from the exact mass identities and one explicit algebraic sign change. The transport reductions and analytic sign estimates are given explicitly, with a table of the finite polynomial inequalities used for the three global signs.

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Contents

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1 Introduction

A function $f : \mathbb{R} \to \mathbb{R}$ is PF_r when every translation-kernel minor

$$\det[f(x_i - y_j)]_{i,j=1}^{k}$$

is nonnegative for $1 \le k \le r$ and increasing node tuples. We call the property strict when the determinant is positive whenever both tuples are strictly increasing. This paper determines the exact finite Pólya-frequency order of the Riemann kernel; see [3] for modern background and Schoenberg's characterization. Related finite-order and Riemann-kernel work is placed in Section 2.

Theorem 1.1 (Exact global PF order)**.** *The exact global Pólya-frequency order of Φ is four. More precisely:*

1. *for every $1 \le k \le 4$ and strictly increasing real tuples $x_1 < \cdots < x_k$ and $y_1 < \cdots < y_k$,*

$$\det[\Phi(x_i - y_j)]_{i,j=1}^{k} > 0.$$

2. *with $h = 211/2000$,*

$$\det[\Phi((i-j)h)]_{i,j=0}^{4} < 0.$$

The two theorem inputs are deliberately separated. The positive input is the universal strict-order theorem through four. The negative input is the direct rational order-five witness in part (2), including proved strict ordering of its nodes and the signed Toeplitz-to-translation-matrix identification.

The proof has two one-variable inputs and one exact transport mechanism. First, global inequalities $q > 0$ and $F_2 > 0$ imply strict PF_3 and positivity of a three-point functional Λ. Second, the fully confluent order-four invariant $\mathcal{C}_4$ is globally positive. Exact quotient identities reduce strict PF_4 to the monotonicity of a scalar three-point function Ψ. On the actual image of an increasing curvature coordinate, the derivative of Ψ becomes a positive integral of $\mathcal{C}_4$ against a deterministic closed cumulative gap.

The finite part of the positive input consists of exact rational Bernstein coefficients on fixed algebraic domains, followed by analytic bounds for all remaining theta modes; Appendix B lists the transformed target polynomials, domains, degrees, minimum coefficients, derivative envelopes, and the surviving relative margins. The transport algebra is proved in the text, and the order-five calculation uses rational enclosures and elementary series bounds.

2 Relation to earlier work

Schoenberg introduced Pólya-frequency functions and connected translation minors with bilateral Laplace transforms and variation-diminishing convolution operators [1, 3]. Karlin's monograph remains the standard systematic account of total positivity and its composition principles [2]; a modern review of PF functions and their preservers is given by Belton, Guillot, Khare, and Putinar [3].

Finite-order translation kernels have a substantially wider range than the infinite-order class. Khare's characterization of measurable TN_p functions for $p \geq 3$ provides a general finite-order setting [4]. The present proof instead derives a criterion specific to order four, using successive quotient derivatives and a curvature transport identity.

The theta normalization used here is the classical one in de Bruijn's study of the Riemann Fourier kernel [5]. For the same kernel, Dimitrov and Xu relate Wronskians of Fourier and Laplace transforms to correlation kernels and real-zero criteria [6], while Csordas and Varga obtain strict concavity and moment inequalities connected with the Riemann Hypothesis [7]. Grochenig explains the distinct Schoenberg correspondence relevant to the Riemann Hypothesis: the PF function there is recovered from the reciprocal of a Laguerre–Pólya entire function, rather than identified with the original Riemann kernel [8].

Michałowski first published an explicit negative 5×5 Toeplitz minor for this kernel, supported by an interval computation, together with single-configuration positivity checks at orders at most four [9]. Global PF_4 is posed there as Open Problem 1. Theorem 1.1 resolves that problem and supplies an exact rational negative order-five minor, completing the finite-order classification.

3 The kernel and smooth even continuation

Put

$$\vartheta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x}, \qquad \Theta(t) = \vartheta(e^{2t}), \qquad H(t) = e^{t/2}\Theta(t).$$

This is a real-valued definition: summability is first proved directly for every $x > 0$, before any analytic continuation is invoked. The real specialization of Gaussian Poisson summation gives the exact normalization $\vartheta(x) = x^{-1/2}\vartheta(1/x)$, and hence

$$\Theta(-t) = e^t\Theta(t), \qquad H(-t) = H(t).$$

A Jacobi-theta realization supplies a globally analytic function agreeing with this real series. Thus H is real analytic and even. Define

$$\Phi(t) = \frac{1}{2}\left(H''(t) - \frac{1}{4}H(t)\right).$$

This is twice the common de Bruijn kernel normalization; multiplication by a positive constant changes none of the PF signs or strictness statements. On $t \geq 0$, locally uniform polynomial–geometric majorants justify termwise differentiation through order six and give

$$\Phi(t) = \sum_{n \geq 1} \left(4\pi^2 n^4 e^{9t/2} - 6\pi n^2 e^{5t/2} \right) e^{-\pi n^2 e^{2t}}. \tag{3.1}$$

Parity reflects this identity to every real t, equivalently by replacing t on the right by $|t|$. Thus the positive-side formula is the restriction of a real-analytic even function. Smoothness and every derivative used later at the origin are consequences of that analytic realization, not of numerical coverage or of differentiating an absolute-value definition.

For $t \geq 0$, the coefficient of the n-th exponential can be factored as

$$2\pi n^2 e^{5t/2} \big(2\pi n^2 e^{2t} - 3 \big) > 0,$$

because $2\pi > 3$. Hence every term in (3.1) is positive and $\Phi(t) > 0$ on $[0, \infty)$. Evenness gives $\Phi(t) > 0$ on all of $\mathbb{R}$, so the following logarithm is globally defined.

Write

$$\ell = \log \Phi, \qquad s = \ell', \qquad q = -s' = -\ell''.$$

For $a < b$, define

$$A(a, b) = s(a) - s(b), \qquad M(a, b) = \frac{q(b) - q(a)}{A(a, b)}. \tag{3.2}$$

When $q > 0$, s is strictly decreasing and $A(a, b) > 0$.

4 Certified one-variable inputs

Define

$$F_1 = qq'' - (q')^2, \qquad F_2 = q^3 - F_1.$$

For the logarithmic cumulants $c_j = \ell^{(j)}$, set

$$\begin{aligned} &m_0 = 1, \quad m_1 = 0, \quad m_2 = c_2, \quad m_3 = c_3, \\ &m_4 = c_4 + 3c_2^2, \quad m_5 = c_5 + 10c_2c_3, \\ &m_6 = c_6 + 15c_2c_4 + 10c_3^2 + 15c_2^3, \end{aligned}$$

and

$$\mathcal{C}_4(t) = \det[m_{i+j-2}(t)]_{i,j=1}^4. \tag{4.1}$$

Lemma 4.1 (Certified input)**.** *For the kernel Φ,*

$$q(t) > 0, \qquad F_2(t) > 0, \qquad \mathcal{C}_4(t) > 0 \quad (t \in \mathbb{R}).$$

The proof combines exact rational polynomial inequalities with analytic theta-tail bounds. For $u \geq 0$, put

$$x = \pi e^{2u}, \qquad \eta = e^{-3x}, \qquad R_0(z) = 2z - 3,$$

and recursively

$$R_{j+1}(z) = \left(\frac{5}{2} - 2z \right) R_j(z) + 2z R_j'(z).$$

After division by the positive first theta mode, the first two modes give the normalized jet entries

$$a_j(x,\eta) = \frac{R_j(x) + 4\eta R_j(4x)}{R_0(x)}.$$

Exact rational Bernstein coefficients on three fixed algebraic domains prove

$$q_{1,2} > 10, \qquad (F_2)_{1,2} > 1000, \qquad (\mathcal{C}_4)_{1,2} > 50{,}000{,}000.$$

Here the half-line certificates are applied directly to the cleared numerators of the shifted targets $q_{1,2} - 10$, $(F_2)_{1,2} - 1000$, and $(\mathcal{C}_4)_{1,2} - 50{,}000{,}000$. The complete contribution of modes $n \geq 3$ is bounded by one monotone polynomial-geometric estimate. Its changes in the same cleared sign numerators are smaller than the three displayed margins, both on $\pi \leq x \leq 5$ and on $x \geq 5$. Evenness completes the real line. All inequalities, including the derivative envelopes used on the tail, are derived in Appendix B. Exact raw-jet identities remove the positive denominators, and the reflected global jet from Section 3 covers the negative half-line and the origin.

5 Order three and the positive slope functional

For $\xi < m < r$, define

$$\Lambda(\xi; m, r) = A(\xi, r) + M(\xi, m) - M(m, r). \tag{5.1}$$

We give the full estimate used to prove its positivity.

Set $f_0 = q'/q$. Because $s' = -q$,

$$M(a,b) = \frac{\int_a^b q'(t)\,\mathrm{d}t}{\int_a^b q(t)\,\mathrm{d}t} = \frac{\int_a^b f_0(t)q(t)\,\mathrm{d}t}{\int_a^b q(t)\,\mathrm{d}t};$$

thus $M(a,b)$ is a q-weighted mean of f_0. Choose a point $u \in [\xi, m]$ where f_0 is minimal on the left interval and a point $v \in [m, r]$ where it is maximal on the right. The weighted-mean bounds and the ordering of the extremizers are

$$M(\xi, m) \geq f_0(u), \qquad M(m, r) \leq f_0(v), \qquad u \leq m \leq v.$$

Consequently

$$M(m,r) - M(\xi,m) \leq f_0(v) - f_0(u) \leq \int_\xi^r \max(f_0'(t), 0)\,\mathrm{d}t.$$

Since

$$f_0' = \frac{qq'' - (q')^2}{q^2} = \frac{F_1}{q^2}$$

and $A(\xi, r) = \int_\xi^r q(t)\,\mathrm{d}t$, equation (5.1) yields

$$\Lambda(\xi; m, r) \geq \int_\xi^r \left(q - \frac{\max(F_1, 0)}{q^2} \right) \mathrm{d}t \tag{5.2}$$

$$= \int_\xi^r \frac{q^3 - \max(F_1, 0)}{q^2}\,\mathrm{d}t = \int_\xi^r \frac{\min(q^3, F_2)}{q^2}\,\mathrm{d}t > 0. \tag{5.3}$$

The second equality uses $q^3 - \max(F_1, 0) = \min(q^3, q^3 - F_1) = \min(q^3, F_2)$; the last inequality follows from Lemma 4.1 and $q > 0$.

The same quantity is the logarithmic slope derivative in the normalized order-three determinant. Hence (5.3), together with strict log-concavity, proves strict global PF_3. We rederive the needed Wronskian form in the next section, where the quotient integral transfers these confluent signs to every ordered collocation minor.

6 Quotient-Wronskian reduction

Fix $y_1 < y_2 < y_3 < y_4$, put $u_j(t) = \Phi(t - y_j)$, and write $p_j = t - y_j$, so $p_4 < p_3 < p_2 < p_1$. Define

$$v_j = (u_j/u_1)', \qquad w_j = (v_j/v_2)'.$$

These quotients are introduced only after their denominators are known to be positive: $u_1 > 0$ by Section 3, and $v_2 > 0$ by the next display. Later division by w_3 is likewise made only after (6.8) proves $w_3 > 0$. Strict log-concavity gives

$$v_j = \frac{u_j}{u_1} A(p_j, p_1) > 0 \qquad (j \geq 2).$$

The case $k = 2$ of Lemma 6.1 below therefore proves strict order two directly. At the fixed orders used in this paper, successive column division and differentiation give the exact three-stage cascade

$$W(u_1, u_2, u_3) = u_1^3 W(v_2, v_3), \tag{6.1}$$

$$W(u_1, u_2, u_3, u_4) = u_1^4 W(v_2, v_3, v_4), \tag{6.2}$$

$$W(v_2, v_3, v_4) = v_2^3 W(w_3, w_4), \tag{6.3}$$

$$W(w_3, w_4) = w_3^2 \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{w_4}{w_3}\right). \tag{6.4}$$

Consequently

$$W(u_1, u_2, u_3, u_4) = u_1^4 v_2^3 w_3^2 \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{w_4}{w_3}\right). \tag{6.5}$$

These algebraic identities require precisely the stated nonvanishing needed to form the quotients.

Let $\mathcal{T}$ denote simultaneous translation of all point arguments. From (3.2),

$$\mathcal{T} A(a, b) = q(b) - q(a), \qquad \mathcal{T} \log A(a, b) = M(a, b). \tag{6.6}$$

The sign convention in (6.6) will be used below. Since

$$\frac{v_j}{v_2} = \frac{u_j}{u_2} \frac{A(p_j, p_1)}{A(p_2, p_1)},$$

direct differentiation gives

$$g_j := \mathcal{T} \log(v_j/v_2) = A(p_j, p_2) + M(p_j, p_1) - M(p_2, p_1). \tag{6.7}$$

The exact weighted-mean split

$$A(p_j, p_1) M(p_j, p_1) = A(p_j, p_2) M(p_j, p_2) + A(p_2, p_1) M(p_2, p_1)$$

turns (6.7) into

$$g_j = \frac{A(p_j, p_2)}{A(p_j, p_1)} \Lambda(p_j; p_2, p_1) > 0. \tag{6.8}$$

Thus $w_j = (v_j/v_2) g_j > 0$, and (6.1) has the strict order-three sign.

Define

$$\Psi(\xi; m, r) = \Lambda(\xi; m, r) + \mathcal{T} \log \Lambda(\xi; m, r).$$

Because $w_j = (v_j/v_2) g_j$,

$$\mathcal{T} \log(w_4/w_3) = (g_4 + \mathcal{T} \log g_4) - (g_3 + \mathcal{T} \log g_3). \tag{6.9}$$

Equation (6.8) and (6.6) give

$$\mathcal{T}\log g_j = M(p_j,p_2) - M(p_j,p_1) + \mathcal{T}\log\Lambda_j,$$

where $\Lambda_j = \Lambda(p_j;p_2,p_1)$. A second use of the same mean split gives

$$g_j + M(p_j,p_2) - M(p_j,p_1) = \Lambda_j - A(p_2,p_1).$$

The common last term cancels in (6.9), yielding the exact identity

$$\mathcal{T}\log(w_4/w_3) = \Psi(p_4;p_2,p_1) - \Psi(p_3;p_2,p_1). \tag{6.10}$$

Lemma 6.1 (Iterated quotient integral)**.** *Let $f_1,\dots,f_k$ be continuously differentiable on an interval, let $f_1>0$, and put $G_j = f_j/f_1$. For $t_1<\cdots<t_k$,*

$$\det[f_j(t_i)]_{i,j=1}^k = \prod_{i=1}^k f_1(t_i)\int_{t_1}^{t_2}\cdots\int_{t_{k-1}}^{t_k}\det[G'_{j+1}(\tau_i)]_{i,j=1}^{k-1}\,\mathrm{d}\tau_1\cdots\mathrm{d}\tau_{k-1}. \tag{6.11}$$

The identity may be applied repeatedly whenever the first member of the new derivative family is positive.

Proof. Factor $f_1(t_i)$ from row i. The first column is then one. Replace successive rows, from bottom to top, by their difference with the preceding row. Each remaining entry is $G_j(t_{i+1}) - G_j(t_i) = \int_{t_i}^{t_{i+1}} G'_j(\tau)\,\mathrm{d}\tau$. Determinant multilinearity gives (6.11). Since $\tau_i\in[t_i,t_{i+1}]$, the integration nodes remain ordered, so the same argument applies to the derivative determinant. □

Lemma 6.2 (Confluent divided-difference limit)**.** *Let $f\in C^{2k-2}$, let $V(a)=\prod_{0\le i<j<k}(a_j-a_i)$, and let a,b be fixed strictly increasing k-tuples. Then*

$$\lim_{h\downarrow 0}\frac{\det[f(z+h(a_i-b_j))]_{i,j=0}^{k-1}}{h^{k(k-1)}V(a)V(b)} = \frac{(-1)^{k(k-1)/2}}{\prod_{r=0}^{k-1}(r!)^2}\det[f^{(i+j)}(z)]_{i,j=0}^{k-1}. \tag{6.12}$$

If only the row nodes coalesce while $y_0<\cdots<y_{k-1}$ remain fixed, then

$$\lim_{h\downarrow 0}\frac{\det[f(z+ha_i-y_j)]_{i,j=0}^{k-1}}{h^{k(k-1)/2}V(a)} = \frac{\det[f^{(i)}(z-y_j)]_{i,j=0}^{k-1}}{\prod_{r=0}^{k-1} r!}. \tag{6.13}$$

Proof. Apply successive Newton divided-difference row operations, whose divisors are the positive numbers $h(a_j-a_i)$, and let $h\downarrow 0$. This gives (6.13). Applying the same operation to the columns gives derivatives with respect to $-y$, hence the factor $(-1)^{0+\cdots+(k-1)}$, and proves (6.12). This records the factorial normalization and does not assume a nonzero leading coefficient. □

Proposition 6.3 (Three-point equivalence)**.** *Under the strict lower-order positivity already proved, global* PF_4 *is equivalent to*

$$\partial_\xi\Psi(\xi;m,r)\le 0 \qquad (\xi<m<r).$$

Strict inequality implies strict global PF_4*.*

Proof. If $\partial_\xi \Psi \le 0$, then $p_4 < p_3$ makes the right side of (6.10) nonnegative. All prefactors in (6.5) are positive. The Wronskian is therefore nonnegative, and is positive in the strict case.

For clarity, applying Lemma 6.1 three times to the order-four minor gives, with $I_i = [t_i, t_{i+1}]$,

$$\begin{aligned}\det[u_j(t_i)]_{i,j=1}^4 = \prod_{i=1}^4 u_1(t_i) \int_{I_1\times I_2\times I_3} \prod_{i=1}^3 v_2(\tau_i) \\ \times \int_{\tau_1}^{\tau_2}\int_{\tau_2}^{\tau_3} w_3(\sigma_1)w_3(\sigma_2)\int_{\sigma_1}^{\sigma_2}\left(\frac{w_4}{w_3}\right)'(\rho)\,\mathrm{d}\rho\,\mathrm{d}\sigma_2\,\mathrm{d}\sigma_1\,\mathrm{d}\boldsymbol{\tau}.\end{aligned}$$

Every factor is positive in the strict case, and every integration interval has positive length. This proves positivity of every ordered collocation minor, not only of the confluent Wronskian.

Conversely, assume global PF_4. Fix arbitrary $\xi_1 < \xi_2 < m < r$, choose any t, and choose the columns so that $(p_4, p_3, p_2, p_1) = (\xi_1, \xi_2, m, r)$ at t; explicitly, $y_j = t - p_j$, which preserves $y_1 < \cdots < y_4$. Coalesce arbitrary strictly ordered row nodes at t. Equation (6.13) and global PF_4 make the resulting translate Wronskian nonnegative. Equations (6.5) and (6.10) therefore give

$$\Psi(\xi_1; m, r) \ge \Psi(\xi_2; m, r).$$

Because both the pair $\xi_1 < \xi_2$ and the pair $m < r$ were arbitrary, $\Psi(\cdot; m, r)$ is nonincreasing for every admissible m, r. Smoothness makes this equivalent to the stated differential inequality. □

7 Curvature coordinate and sign bridge

Since $q > 0$,

$$y(t) = -s(t), \qquad Q(y(t)) = q(t)$$

defines a global strictly increasing coordinate on its image, with $\mathrm{d}y/\mathrm{d}t = q(t) = Q(y) > 0$. Primes in this and the next three sections mean y-derivatives. Put

$$\rho = 1 - Q'' = \frac{F_2}{q^3} > 0, \qquad \kappa = 2 - Q'' = 1 + \rho > 1.$$

For $\xi < m < r$, write

$$p = y(\xi), \quad z = y(m), \quad w = y(r), \quad L = z - p, \quad R = w - z.$$

Because $A(a,b) = y(b) - y(a)$ and $M(a,b) = Q[y(a), y(b)]$,

$$\Lambda = (w - p) + Q[p, z] - Q[z, w]. \tag{7.1}$$

Two integrations of $\kappa = 2 - Q''$ give

$$\Lambda = \frac{1}{L}\int_p^z (y - p)\kappa(y)\,\mathrm{d}y + \frac{1}{R}\int_z^w (w - y)\kappa(y)\,\mathrm{d}y, \tag{7.2}$$

$$\delta := -\partial_p \Lambda = \frac{1}{L^2}\int_p^z (z - y)\kappa(y)\,\mathrm{d}y > 0. \tag{7.3}$$

Simultaneous translation becomes

$$\mathcal{T} = Q(p)\partial_p + Q(z)\partial_z + Q(w)\partial_w,$$

and $\partial_\xi = Q(p)\partial_p$. Thus $\Lambda_\xi = -Q(p)\delta$.

Define

$$\mathcal{N} = \delta\Lambda^2 + Q'(p)\delta\Lambda + \Lambda\mathcal{T}\delta - \delta\mathcal{T}\Lambda. \tag{7.4}$$

Differentiating $\Psi = \Lambda + \mathcal{T}\log\Lambda$, commuting simultaneous translation with ∂_ξ, and collecting over Λ^2 gives

$$\partial_\xi\Psi = -\frac{Q(p)}{\Lambda^2}\mathcal{N}. \tag{7.5}$$

Equivalently,

$$\frac{\mathcal{N}}{\delta\Lambda} = K := \Lambda + Q'(p) + \mathcal{T}\log\delta - \mathcal{T}\log\Lambda. \tag{7.6}$$

It remains to prove $K > 0$.

All objects in this coordinate are evaluated on the image $y(\mathbb{R})$, where the inverse coordinate is well defined. The displayed ρ, κ signs hold at every such point, and simultaneous translation of the original endpoints is exactly the operator $\mathcal{T}$ above.

8 The confluent invariant in curvature form

Define

$$D = 3(\kappa - 1) - \{Q(\log\kappa)'\}'. \tag{8.1}$$

Lemma 8.1 (Confluent-to-curvature identity)**.** *The determinant* (4.1) *satisfies*

$$\mathcal{C}_4 = Q^6\kappa^2 D. \tag{8.2}$$

Consequently $D > 0$ on $\mathbb{R}$.

Proof. The change of variable gives $\mathrm{d}/\mathrm{d}t = Q\,\mathrm{d}/\mathrm{d}y$ and $c_2 = -Q$. Recursively applying this operator produces the five cumulants listed in Appendix A. Substitution into the Schur-complement form of (4.1) factors out Q^6. The remaining factor is

$$\kappa^2\left(3(\kappa - 1) - \{Q(\log\kappa)'\}'\right),$$

as shown by the common expanded numerator in Appendix A. This proves (8.2) without a numerical assumption. Lemma 4.1, $Q > 0$, and $\kappa > 0$ then imply $D > 0$. $\square$

The determinant definition is primary: it is the doubly confluent fourth-order translation minor after division by the positive row and column Vandermonde factors and by Φ^4. The explicit thirteen-term expansion is included in Appendix A for independent checking of the certificate input.

9 Exact endpoint transport identity

Write the three normalized triangular densities from (7.2) and (7.3) as

$$\mu_L(y) = \frac{(z-y)\kappa(y)}{L^2\delta} \quad (p \le y \le z), \tag{9.1}$$

$$\nu_L(y) = \frac{(y-p)\kappa(y)}{L\Lambda} \quad (p \le y \le z),$$

$$\nu_R(y) = \frac{(w-y)\kappa(y)}{R\Lambda} \quad (z \le y \le w). \tag{9.2}$$

The triangular identities give

$$\int_p^z \mu_L = 1, \qquad \int_p^z \nu_L + \int_z^w \nu_R = 1. \tag{9.3}$$

No measure theory is needed below; these are normalized continuous weights on compact intervals.

Set

$$A_0(y) = 3(y - Q'(y)) - Q(y)\frac{\kappa'(y)}{\kappa(y)}.$$

Then $A_0' = D$.

Lemma 9.1 (Endpoint transport cancellation)**.** *With K as in* (7.6)*, put*

$$\mathcal{E} = \frac{-I_L/L + I_R/R}{\Lambda} + \frac{L\mathcal{H}(p) - I_L}{L^2\delta}, \tag{9.4}$$

where $I_L = \mathcal{J}(z) - \mathcal{J}(p)$ and $I_R = \mathcal{J}(w) - \mathcal{J}(z)$, with $\mathcal{H}, \mathcal{J}$ defined below. Then

$$K = \mathcal{E}. \tag{9.5}$$

Proof. Two elementary primitives carry the complete cancellation:

$$\kappa A_0 = \mathcal{H}', \qquad \mathcal{H} = 3y^2 - 3Q - 3yQ' + (Q')^2 + QQ'', \tag{9.6}$$

$$\mathcal{H} = \mathcal{J}', \qquad \mathcal{J} = y^3 - 3yQ + QQ'. \tag{9.7}$$

Both identities follow by one differentiation. Integrating the three linear weights once by parts gives

$$\int_p^z A_0\nu_L + \int_z^w A_0\nu_R - \int_p^z A_0\mu_L = \mathcal{E}. \tag{9.8}$$

On the other hand, differentiation of the triangular integrals along $\mathcal{T}$ gives

$$\mathcal{T}\Lambda = U(z,w) - U(p,z), \quad \mathcal{T}\delta = \frac{U(p,z) - (Q(z) - Q(p))\delta - Q(p)\kappa(p)}{L},$$

where

$$U(c,d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d-c} + \frac{(Q(d) - Q(c))^2}{(d-c)^2}. \tag{9.9}$$

Substitution in (7.6) yields

$$K = \Lambda + Q'(p) - Q[p,z] + \frac{U(p,z) - U(z,w)}{\Lambda} + \frac{U(p,z) - Q(p)\kappa(p)}{L\delta}. \tag{9.10}$$

The endpoint identity equating (9.4) and (9.10), with every substitution exposed, is Appendix C. It is an identity in the endpoint positions and the values of Q, Q', Q''. □

10 The positive closed coordinate gap

Define a deterministic piecewise function on $[p,w]$ by

$$\Delta(y) = \begin{cases} \dfrac{-(z-y)^2 - (z-y)Q'(y) - Q(y) + (z-p)^2 + (z-p)Q'(p) + Q(p)}{L^2\delta} \\ \qquad - \dfrac{(y-p)^2 - (y-p)Q'(y) + Q(y) - Q(p)}{L\Lambda}, & p \le y \le z, \\ \dfrac{(w-y)^2 + (w-y)Q'(y) + Q(y) - Q(w)}{R\Lambda}, & z < y \le w. \end{cases} \tag{10.1}$$

The fundamental theorem of calculus and $\kappa = 2 - Q''$ identify these closed expressions with

$$\Delta(y) = \begin{cases} \displaystyle\int_p^y (\mu_L(t) - \nu_L(t))\,\mathrm{d}t, & p \le y \le z, \\ \displaystyle\int_y^w \nu_R(t)\,\mathrm{d}t, & z < y \le w. \end{cases} \tag{10.2}$$

The two branches agree at z by (9.3).

Lemma 10.1 (Strict closed gap)**.** *The function Δ is continuous on $[p,w]$, satisfies $\Delta(p) = \Delta(w) = 0$, and*

$$\Delta(y) > 0 \qquad (p < y < w).$$

Proof. The endpoints and continuity follow from (10.1), (10.2), and the exact mass identities. On (p,z), the difference of the positive left densities changes sign at the explicit strict convex combination

$$y_* = \frac{\Lambda z + L\delta p}{\Lambda + L\delta} \in (p,z).$$

Indeed, clearing the positive common factor $\kappa(y)$ and the positive denominators gives

$$\mu_L(y) - \nu_L(y) > 0 \iff y < y_*, \qquad \mu_L(y) - \nu_L(y) < 0 \iff y > y_*.$$

Thus $\Delta(y) > 0$ for $p < y \le y_*$. For $y_* \le y \le z$, subtract the two identities in (9.3) and split at y to obtain

$$\Delta(y) = \int_y^z (\nu_L - \mu_L)\,\mathrm{d}t + \int_z^w \nu_R\,\mathrm{d}t > 0.$$

The first integral is nonnegative and the second is strictly positive because $z < w$, $\kappa > 0$, and $\Lambda > 0$. On (z,w), positivity follows directly from the positive tail integral in (10.2). $\square$

On the left branch $\Delta' = \mu_L - \nu_L$, and on the right branch $\Delta' = -\nu_R$. Direct integration by parts on $[p,z]$ and $[z,w]$, using $A_0' = D$, the zero endpoint values, and the equality of the two branches at z, gives

$$\begin{aligned} \int_p^w \Delta(y) D(y)\,\mathrm{d}y &= \int_p^z A_0\nu_L + \int_z^w A_0\nu_R - \int_p^z A_0\mu_L \\ &= \mathcal{E} = K. \end{aligned} \tag{10.3}$$

Combining (7.6), (8.2), and (10.3) gives the central identity

$$\mathcal{N} = \delta\Lambda \int_p^w \Delta(y) \frac{\mathcal{C}_4(y)}{Q(y)^6 \kappa(y)^2}\,\mathrm{d}y > 0. \tag{10.4}$$

Every factor in the interior is strictly positive.

11 Completion and exact order

Equations (7.5) and (10.4) imply

$$\partial_\xi \Psi(\xi; m, r) < 0 \qquad (\xi < m < r).$$

Proposition 6.3 therefore proves part (1) of Theorem 1.1.

We now prove directly that the classification stops at order four. Let $h = 211/2000$. Exact rational theta enclosures give

$$-1.08 \cdot 10^{-7} < \det[\Phi((i-j)h)]_{i,j=0}^4 < -1.05 \cdot 10^{-7}. \tag{11.1}$$

Theorem 11.1. *The Riemann kernel is not* PF_5*.*

Proof. The nodes $0, h, 2h, 3h, 4h$ are strictly increasing. Evenness identifies the translation matrix with the symmetric Toeplitz matrix determined by the five kernel values $\Phi(0), \Phi(h), \dots, \Phi(4h)$. Each value is enclosed by exact Taylor bounds for the retained theta modes and the infinite-tail theorem; a rational determinant parity factorization then gives (11.1). This is a negative translation minor at distinct nodes. □

Equation (11.1) proves part (2) of Theorem 1.1. Together with the strict minors through order four, this completes the exact-order classification.

Remark 11.2 (Riemann Hypothesis boundary). Schoenberg's transform theorem says that the bilateral Laplace transform of a PF_∞ function is the reciprocal of a Laguerre–Pólya entire function on its strip of convergence [1, 3]. The Fourier transform of Φ, up to the standard normalization, is the Riemann Ξ-function, which has known real zeros; hence Φ is unconditionally not PF_∞. The Schoenberg formulation related to the Riemann Hypothesis instead asks for the inverse transform of the reciprocal of Ξ to be a Pólya-frequency function [8]. Thus the finite-order classification proved here is a different statement.

12 Verification

The verification boundary is the theorem `PF4.globalRiemannKernel_pfOrderExactly_four` in the project `proof/formal`. It has two immediate mathematical inputs: `PF4.globalRiemannKernel_strictPFUpTo_four`, which quantifies over every order $1 \le k \le 4$ and every pair of strictly increasing real node tuples, and `PF4.globalRiemannKernel_orderFive_translationMinor_neg`, which proves the negative minor at $h = 211/2000$. The former depends on the analytic theta construction, the global $q, F_2, \mathcal{C}_4$ signs, and the quotient–transport argument developed in Sections 3–10.

The following criteria define the formal verification claim and make it reproducible.

1. **Pinned environment.** Lean `v4.32.0` is fixed in `lean-toolchain`; mathlib `v4.32.0` is fixed in `lake-manifest.json`. The resolved mathlib commit is

   `81a5d257c8e410db227a6665ed08f64fea08e997`.

2. **Complete build.** From `proof/formal`, the command

   ```
   lake build
   ```

   elaborates every imported declaration and checks every generated proof term with the Lean kernel. The recorded release build completes all 3737 jobs.

3. **Axiom-closure audit.** The command

   ```
   lake env lean PF4/Audit.lean
   ```

   runs `#print axioms` on the final theorem, both immediate inputs, and the target-facing declarations in their proof chain. For every audited target the printed dependency set is exactly

   {`propext`, `Classical.choice`, `Quot.sound`}.

These are Lean's standard logical axioms. A declaration containing `sorry` or `admit` elaborates with the axiom `sorryAx`; consequently such a gap anywhere in an audited target's dependency closure would appear in this output. The observed output contains no `sorryAx` and no project-declared axiom.

4. **Exact sign decisions.** The compact sign certificates, theta-tail bounds, and the order-five determinant enclosure enter the theorem as exact integer or rational inequalities. Floating-point values do not discharge a proposition in the final dependency closure.

5. **Quantifier coverage.** The strict-order conclusion is a single universal theorem over arbitrary real node tuples; it is not a finite node sample. The negative order-five conclusion includes the strict ordering of the five rational nodes, the translation-matrix identity, and the determinant sign.

Two independent Python replays expose the finite arithmetic printed in the paper. From the repository root,

Claim	Command
Global $q, F_2, \mathcal{C}_4$ signs	`python3 scripts/verify_riemann_signs_exact.py`
Negative minor at $h = 211/2000$	`python3 scripts/verify_pf5_witness_exact.py`

Both use outward rational bounds. They are independently inspectable reconstructions of the finite certificates; the universal theorem remains the kernel-checked Lean declaration above. Tool versions, output digests, the PDF digest, and the source-archive digest are recorded in `paper/RELEASE_MANIFEST.md`.

13 Declarations and availability

Author contribution. Joshuah Rainstar conceived and directed the project, developed the proof architecture, verified the exact calculations, and prepared the manuscript.

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Conflict of interest. The author declares no competing interests.

Code and data availability. The public repository is `https://github.com/falseywinchnet/zeta`. The submitted manuscript is preserved by the immutable tag `pf4-paper-v1.0.0`, commit `4a14988093d7afaf05f453f8daf5441c09ba58b3`. The present revised proof is prepared as release `pf4-paper-v2.1.0`. Their distinct status and material proof changes are listed in `paper/REVISION_HISTORY.md`; current source, environment, verification, PDF, and archive digests are listed in `paper/RELEASE_MANIFEST.md`. The repository archive is assigned DOI

`https://doi.org/10.5281/zenodo.21487371`.

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A Confluent and curvature algebra

The central-moment Hankel matrix is

$$\begin{pmatrix} 1 & 0 & m_2 & m_3 \\ 0 & m_2 & m_3 & m_4 \\ m_2 & m_3 & m_4 & m_5 \\ m_3 & m_4 & m_5 & m_6 \end{pmatrix}.$$

Taking the Schur complement of the upper-left entry reduces its determinant to

$$\det \begin{pmatrix} m_2 & m_3 & m_4 \\ m_3 & m_4 - m_2^2 & m_5 - m_2 m_3 \\ m_4 & m_5 - m_2 m_3 & m_6 - m_3^2 \end{pmatrix}.$$

Substitution of the moments from Section 4 and ordinary three-by-three expansion gives

$$\mathcal{C}_4 = 12c_2^6 + 24c_2^4c_4 - 24c_2^3c_3^2 + 2c_2^3c_6 - 12c_2^2c_3c_5 + 7c_2^2c_4^2 \tag{A.1}$$

$$+ 12c_2c_3^2c_4 + c_2c_4c_6 - c_2c_5^2 - 9c_3^4 - c_3^2c_6 + 2c_3c_4c_5 - c_4^3. \tag{A.2}$$

In the curvature coordinate, repeated application of $Q\,\mathrm{d}/\,\mathrm{d}y$ to $c_2 = -Q$ gives

$$\begin{aligned}
c_2 &= -Q,\\
c_3 &= -QQ',\\
c_4 &= -Q\{(Q')^2 + QQ''\},\\
c_5 &= -Q\{(Q')^3 + 4QQ'Q'' + Q^2Q'''\},\\
c_6 &= -Q\{(Q')^4 + 11Q(Q')^2Q'' + 4Q^2(Q'')^2 + 7Q^2Q'Q''' + Q^3Q''''\}.
\end{aligned}$$

Substitution in the preceding polynomial gives

$$\mathcal{C}_4 = -Q^6\mathcal{P},$$

where

$$\begin{aligned}
\mathcal{P} = {} & QQ''Q'''' - Q(Q''')^2 - 2QQ'''' + Q'Q''Q''' - 2Q'Q'''\\
& + 3(Q'')^3 - 15(Q'')^2 + 24Q'' - 12.
\end{aligned}$$

Now $\kappa = 2 - Q''$, and direct differentiation gives

$$\{Q(\log\kappa)'\}' = \frac{Q'\kappa' + Q\kappa''}{\kappa} - \frac{Q(\kappa')^2}{\kappa^2},$$

so, after multiplication by κ^2 and substitution of $\kappa' = -Q'''$, $\kappa'' = -Q''''$,

$$3(\kappa - 1) - \{Q(\log\kappa)'\}' = -\frac{\mathcal{P}}{(Q'' - 2)^2} = -\frac{\mathcal{P}}{\kappa^2}.$$

Therefore $\mathcal{C}_4 = Q^6\kappa^2 D$, proving (8.2) by an exposed common numerator.

B Exact proof of the certified input

This appendix proves Lemma 4.1. The positive-side theta series has j-th derivative

$$2\pi n^2 e^{5u/2 - n^2x} R_j(n^2x), \qquad x = \pi e^{2u},$$

where the polynomials R_j satisfy the recurrence in Section 4. Machin's identity and alternating series give $\pi > 157/50$; hence $R_0(x) = 2x - 3 > 0$.

B.1 Cleared sign polynomials

In addition to a_j, define the normalized remaining modes

$$e_j(x) = \sum_{n\geq 3} \frac{n^2 e^{-(n^2-1)x} R_j(n^2x)}{R_0(x)}, \qquad \widehat{a}_j = a_j + e_j.$$

For an indeterminate raw jet $(\widehat{a}_0, \ldots, \widehat{a}_6)$, put

$$N_q = \widehat{a}_1^2 - \widehat{a}_0\widehat{a}_2,$$

$$
\begin{aligned}
N_{F_2} = & -\widehat{a}_0^4\widehat{a}_2\widehat{a}_4 + \widehat{a}_0^4\widehat{a}_3^2 + \widehat{a}_0^3\widehat{a}_1^2\widehat{a}_4 - 2\widehat{a}_0^3\widehat{a}_1\widehat{a}_2\widehat{a}_3 + 2\widehat{a}_0^3\widehat{a}_2^3 \\
& - 3\widehat{a}_0^2\widehat{a}_1^2\widehat{a}_2^2 + 3\widehat{a}_0\widehat{a}_1^4\widehat{a}_2 - \widehat{a}_1^6,
\end{aligned}
$$

and

$$
N_{\mathcal{C}_4} = \det[\widehat{a}_{i+j}]_{i,j=0}^3.
$$

Direct logarithmic differentiation and the central-moment determinant give

$$
q = \frac{N_q}{\widehat{a}_0^2}, \qquad F_2 = \frac{N_{F_2}}{\widehat{a}_0^6}, \qquad \mathcal{C}_4 = \frac{N_{\mathcal{C}_4}}{\widehat{a}_0^4}. \tag{B.1}
$$

The thirteen-term cumulant expansion obtained from the last identity agrees with (A.1); this is an exact polynomial identity.

For orientation, retaining only the first mode gives

$$
q_1 = \frac{4x(4x^2 - 12x + 15)}{(2x-3)^2},
$$

$$
(F_2)_1 = \frac{64x^3(64x^6 - 576x^5 + 2448x^4 - 6432x^3 + 10044x^2 - 8100x + 2295)}{(2x-3)^6},
$$

$$
(\mathcal{C}_4)_1 = \frac{49152x^6(16x^4 - 96x^3 + 360x^2 - 840x + 945)}{(2x-3)^4}.
$$

After $x = 5 + z$, every numerator has positive coefficients. The second mode is retained exactly because it supplies the larger margin near the origin.

B.2 Exact two-mode margin

Rational Taylor bounds for the exponential imply

$$
0 < \eta < \frac{1}{12000} \quad (x \geq \pi),
$$

and

$$
\frac{1}{23000} < \eta < \frac{1}{12000} \quad \left(\frac{157}{50} \leq x \leq \frac{10}{3}\right).
$$

The close endpoints are entirely rational checks. Namely,

$$
\sum_{k=0}^{16} \frac{(471/50)^k}{k!} > 12000, \qquad \sum_{k=0}^{21} \frac{10^k}{k!} > 22000,
$$

and the geometric bound for the tail after degree 13 gives

$$
e^{10} < \sum_{k=0}^{13} \frac{10^k}{k!} \frac{10^{14}/14!}{1 - 10/15} = \frac{3480060751}{154791} < 23000.
$$

These prove, respectively, the upper η-cap, the cap $e^{-10} < 1/22000$, and the lower bound $e^{-10} > 1/23000$. Likewise $\sum_{k=0}^{21} 15^k/k! > 3{,}000{,}000$, which gives $\eta < 1/3{,}000{,}000$ for $x \geq 5$. We use the following elementary certificate lemma. If $p(s,t) = \sum p_{ij} s^i t^j$ has bidegree (d, e), its Bernstein coefficients on the unit square are

$$
\beta_{k\ell} = \sum_{i \leq k,\, j \leq \ell} p_{ij} \frac{\binom{k}{i}}{\binom{d}{i}} \frac{\binom{\ell}{j}}{\binom{e}{j}}.
$$

The power-to-Bernstein identity proves that $\beta_{k\ell} > 0$ for every k, ℓ implies $p > 0$ on the square.

Apply this lemma after affine rational changes of variables. For q, one rectangle ends at $x = 10/3$, followed by a positive base polynomial minus exponentially decreasing corrections. For F_2, dependent bounds on the first rectangle are followed by a rectangle through $x = 5$ and a positive half-strip expansion. For $\mathcal{C}_4$, the first rectangle is followed by two negative odd-power corrections whose ratios to the positive base decrease. The derivative of each ratio has a numerator with positive coefficients after the indicated shift. The 361 resulting rational coefficient inequalities give

$$q_{1,2} > 10, \qquad (F_2)_{1,2} > 1000, \qquad (\mathcal{C}_4)_{1,2} > 50{,}000{,}000. \tag{B.2}$$

This is a finite algebraic proof on fixed domains.

Here is the precise connection between those coefficient checks and the quantitative margins. Evaluate the raw polynomials at the two-mode jet and set

$$\mathcal{Q} = N_q - 10a_0^2, \qquad \mathcal{F} = N_{F_2} - 1000a_0^6, \qquad \mathcal{C} = N_{\mathcal{C}_4} - 50{,}000{,}000a_0^4.$$

The rows labelled Q, F, C below certify the cleared numerators of $\mathcal{Q}, \mathcal{F}, \mathcal{C}$, respectively. Thus their positivity is exactly, rather than merely qualitatively, the three inequalities in (B.2). For a ratio row, write the shifted target as $B_0(x) + \sum_{j\geq 1} \eta^j B_j(x)$, where B_0 has positive shifted coefficients. Whenever B_j is adverse, put $D_j = -B_j$. Positivity of

$$3jD_jB_0 - D_j'B_0 + D_jB_0'$$

after the same shift proves that $e^{-3jx}D_j(x)/B_0(x)$ decreases. The displayed endpoint ratio is the sum of all such adverse ratios at the left endpoint; being less than one proves the target positive on the entire half-line. Coefficientwise nonnegative pieces only increase it.

For reference, the complete finite certificate is summarized below. Write $\mathcal{A} = (2x-3)+4(8x-3)\eta$, which is positive on every listed domain. Rows Q_0, F_0, F_1, C_0 are Bernstein certificates after the indicated affine change to the unit square; the remaining rows use coefficient positivity after a half-line shift. In the two "ratio" rows, the listed fraction is the sum of the adverse coefficient ratios and is strictly less than one.

ID	Variables and domain	Degree	Positive denominator	Exact lower datum
Q_0	$157/50 \leq x \leq 10/3,\ 0 \leq \eta \leq 1/12000$	$(4,2)$	$\mathcal{A}^2$	6613625187767/70312500000
Q_∞	$x = 10/3 + z,\ 0 \leq \eta \leq 1/12000$	$(4,2)$	$\mathcal{A}^2$	ratio 47333/505500
F_0	$157/50 \leq x \leq 10/3,\ 1/23000 \leq \eta \leq 1/12000$	$(12,6)$	$\mathcal{A}^6$	minimum > 1610047
F_1	$10/3 \leq x \leq 5,\ 0 \leq \eta \leq 1/22000$	$(12,6)$	$\mathcal{A}^6$	29124973000/19683
F_∞	$x = 5 + z,\ \eta = v/3000000,\ 0 \leq v \leq 1$	$(12,6)$	$\mathcal{A}^6$	432/19073486328125
C_0	$157/50 \leq x \leq 10/3,\ 0 \leq \eta \leq 1/12000$	$(14,4)$	$\mathcal{A}^4$	minimum > 9546672947
C_∞	$x = 10/3 + z,\ 0 \leq \eta \leq 1/12000$	$(14,4)$	$\mathcal{A}^4$	ratio 65989337396066791/77165720550000000

These seven checks comprise 361 exact rational inequalities. The derivative envelopes used below contribute 140 further Bernstein coefficients, of bidegrees $(j+1, 1)$, $0 \leq j \leq 6$; their smallest coefficient is 44861/31250. Thus the compact domains and the differentiated tail majorants belong to the same exact certificate.

B.3 All later modes

For direct reconstruction of the derivative bounds, put $S_j(t) = 2^j(-1)^j R_j(t)$. The recurrence gives

$$
\begin{aligned}
S_0 &= 2t - 3,\\
S_1 &= 8t^2 - 30t + 15,\\
S_2 &= 32t^3 - 224t^2 + 330t - 75,\\
S_3 &= 128t^4 - 1440t^3 + 4232t^2 - 3270t + 375,\\
S_4 &= 512t^5 - 8448t^4 + 41408t^3 - 68096t^2 + 30930t - 1875,\\
S_5 &= 2048t^6 - 46592t^5 + 343040t^4 - 976320t^3 + 1008968t^2 - 285870t + 9375,\\
S_6 &= 8192t^7 - 245760t^6 + 2536960t^5 - 11109120t^4 + 20633312t^3 - 14260064t^2 + 2610330t - 46875.
\end{aligned}
$$

For $t \geq 27$, shifting $t = 27 + z$ gives positive coefficients in $(-1)^j R_j(t)$ and in $2^{j+1}t^{j+1} - (-1)^j R_j(t)$. Therefore

$$0 < (-1)^j R_j(t) \leq 2^{j+1}t^{j+1}, \qquad 0 \leq j \leq 6. \tag{B.3}$$

The derivative numerator of the normalized $n = 3$ term is another positive shifted polynomial, so its absolute value decreases for $x \geq 3$. For $n \geq 4$, successive majorants have ratio at most

$$\left(\frac{5}{4}\right)^{16} e^{-27} < 10^{-9}.$$

Rational Taylor lower bounds for e^{24}, e^{27}, e^{45} then yield, on $\pi \leq x \leq 5$,

$$(|e_0|, \ldots, |e_6|) < (7\cdot 10^{-9}, 4\cdot 10^{-7}, 2\cdot 10^{-4}, 7\cdot 10^{-4}, 3\cdot 10^{-2}, 1.1, 41). \tag{B.4}$$

Indeed, the normalized $n = 3$ term decreases on $x \geq 3$, and $e^{24} > 25{,}000{,}000{,}000$, so it is bounded by $3|R_j(27)|/25{,}000{,}000{,}000$. The sum of the $n \geq 4$ majorants is at most

$$\frac{2^{j+2}3^j 4^{2j+4}}{3 \cdot 10^{19}}.$$

The data entering these two expressions are

j	$\lvert R_j(27)\rvert$	chosen upper bound for $\lvert e_j\rvert$
0	51	$7 \cdot 10^{-9}$
1	5037/2	$4 \cdot 10^{-7}$
2	475395/4	$2 \cdot 10^{-4}$
3	42678141/8	$7 \cdot 10^{-4}$
4	3623251731/16	$3 \cdot 10^{-2}$
5	288709329165/32	11/10
6	21373315648611/64	41

For each row the later-mode expression is less than one thousandth of the $n = 3$ expression, and their sum is strictly below the chosen bound. This derives every constant in (B.4) from the printed recurrence and three rational exponential inequalities. The same Bernstein lemma gives

$$(|a_0|, \ldots, |a_6|) < (2, 5, 20, 200, 2000, 60000, 1000000).$$

For any one of the printed raw polynomials $N = \sum_\alpha c_\alpha X^\alpha$, the perturbation estimate used here is the explicit finite sum

$$|N(a+e) - N(a)| \le \sum_\alpha |c_\alpha| \left\{ \prod_{j=0}^{6} (A_j + E_j)^{\alpha_j} - \prod_{j=0}^{6} A_j^{\alpha_j} \right\}, \tag{B.5}$$

whenever $|a_j| \le A_j$ and $|e_j| \le E_j$. Substitution of the two displayed seven-tuples into (B.5) gives

$$|\Delta N_q| < 0.000405, \quad |\Delta N_{F_2}| < 37.959, \quad |\Delta N_{\mathcal{C}_4}| < 31{,}549{,}532. \tag{B.6}$$

For $x \ge 5$, (B.3) gives

$$|a_j| \le 2^{j+2} x^j, \qquad |e_j| \le 2^{j+2} 3^{2j+4} x^j e^{-8x}.$$

The raw weights of $N_q, N_{F_2}, N_{\mathcal{C}_4}$ are respectively $2, 6, 12$. Thus every perturbation majorant is a positive multiple of $x^w e^{-8x}$, or a faster-decaying term, with $w \le 12$; it decreases for $x \ge 5$. In (B.5) take

$$A_j = 2^{j+2} 5^j, \qquad E_j = \frac{2^{j+1} 3^{2j+4} 5^j}{10^{17}}.$$

The latter follows from the displayed tail bound and $\sum_{k=0}^{47} 40^k/k! > 2 \cdot 10^{17}$. Direct substitution gives

$$|\Delta N_q| < 6.49 \cdot 10^{-11}, \quad |\Delta N_{F_2}| < 0.03, \quad |\Delta N_{\mathcal{C}_4}| < 418{,}287. \tag{B.7}$$

To compare like quantities, note that

$$a_0 = 1 + 4\eta \frac{R_0(4x)}{R_0(x)} > 1,$$

and, by homogeneity,

$$N_{q,1,2} = q_{1,2} a_0^2, \qquad N_{F_2,1,2} = (F_2)_{1,2} a_0^6, \qquad N_{\mathcal{C}_4,1,2} = (\mathcal{C}_4)_{1,2} a_0^4.$$

Consequently (B.2) supplies the same respective lower margins 10, 1000, and 50,000,000 for the cleared numerators. Both perturbation bounds (B.6) and (B.7) are strictly smaller. Their scale relative to the corresponding cleared-numerator margin is recorded here:

target	surviving fraction on $\pi \le x \le 5$	surviving fraction on $x \ge 5$
q	> 0.9999595	> 0.99999999999351
F_2	> 0.962041	> 0.99997
$\mathcal{C}_4$	> 0.36900936	> 0.99163426

Thus even the least relative margin, the compact $\mathcal{C}_4$ bound, retains more than 36.9 percent of its certified two-mode numerator margin. Finally $\widehat{a}_0 = a_0 + e_0 > 0$, so (B.1) proves all three signs for $u \ge 0$. Evenness proves Lemma 4.1.

C Endpoint algebra for the transport cancellation

This appendix records the algebra suppressed by the phrase "simplifying the endpoint expression." Put

$$z = p + L, \qquad w = z + R, \qquad a = \frac{Q(z) - Q(p)}{L}, \qquad b = \frac{Q(w) - Q(z)}{R}.$$

Direct integration of $2 - Q''$ gives

$$\delta = \frac{L + Q'(p) - a}{L}, \qquad \Lambda = L + R + a - b. \tag{C.1}$$

Define

$$\mathcal{J}_y = y^3 - 3yQ(y) + Q(y)Q'(y),$$
$$\mathcal{H}_p = 3p^2 - 3Q(p) - 3pQ'(p) + Q'(p)^2 + Q(p)Q''(p),$$

and, for endpoint data,

$$U(c,d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d - c} + \frac{(Q(d) - Q(c))^2}{(d-c)^2}.$$

With $I_L = \mathcal{J}_z - \mathcal{J}_p$ and $I_R = \mathcal{J}_w - \mathcal{J}_z$, the identity needed in Lemma 9.1 is exactly

$$\begin{aligned} &\frac{-(\mathcal{J}_z - \mathcal{J}_p)/L + (\mathcal{J}_w - \mathcal{J}_z)/R}{\Lambda} + \frac{L\mathcal{H}_p - (\mathcal{J}_z - \mathcal{J}_p)}{L^2\delta} \\ &\quad = \Lambda + Q'(p) - a + \frac{U(p,z) - U(z,w)}{\Lambda} + \frac{U(p,z) - Q(p)(2 - Q''(p))}{L\delta}. \end{aligned}$$

To check it, substitute (C.1), replace z by $p + L$ and w by $p + L + R$, and multiply by the common denominator $L^2R\delta\Lambda$. The terms containing $Q'(w)$, then $Q'(z)$, then $Q''(p)$ cancel pairwise; the remaining endpoint values and position terms cancel after inserting $a = (Q(z) - Q(p))/L$ and $b = (Q(w) - Q(z))/R$. No differential equation for Q is used, so the identity holds for arbitrary smooth Q. Equivalently, regard the endpoint jets $Q(p), Q(z), Q(w), Q'(p), Q'(z), Q'(w), Q''(p)$ and p, L, R as independent indeterminates subject only to the displayed definitions of a, b, δ, and Λ. Clearing $L^2R\delta\Lambda$ first and then expanding gives the zero polynomial. Expanding the uncleared rational expression need not expose this cancellation.